

Technology #114

Implement a comprehensive monitoring system
that combines acoustic sensing to detect lead
dislodgement with thermal monitoring to
detect tissue interface changes

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

WHY DOES IT MATTER?

Documented recurring problem 'adverse_event_without_identified_device_' for pacemaker lead: 16 MAUDE reports across 7 distinct devices (20220222 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 4 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, lead, maintain, matching, modes, must, problem, producing, recurring, report

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in acoustic. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Compare complication rates in patients with standard pacemaker leads versus those with the hybrid acoustic-thermal monitoring system over.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: impedance-contrast detection threshold in phantom. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium; alpha is frequency- and medium-dependent; two-way path doubles the exponent; oblique incidence and attenuation modify detectability; AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.; AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.; AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.; Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A0 (-), alpha (1/m), d (m), Z1 (Rayl), Z2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: lambda (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f, Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: A lifecycle-based risk management framework that integrates complication prevention across the entire continuum of pacemaker therapy

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'adverse_event_without_identified_device_' for pacemaker lead: 16 MAUDE reports across 7 distinct devices (20220222 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 4 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, lead, maintain, matching, modes, must, problem, producing, recurring, report

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: acoustic

Disciplines: acoustics, transducer engineering, signal processing, materials science

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

Subsystems:

?: transducer element + matching layer - ?

?: excitation / receive electronics - ?

?: signal chain (detection criterion) - ?

?: coupling path to target medium - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

Key Physics: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: CONDITIONAL; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: NOT ESTABLISHED.

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous lossy medium; single frequency. Assumptions: alpha is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: NOT ESTABLISHED.

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Assumptions:

- sound speed is medium- and temperature-dependent — source the value for the actual medium
- alpha is frequency- and medium-dependent; two-way path doubles the exponent
- oblique incidence and attenuation modify detectability
- AC-002 (Wave speed relation) is applicable only under: non-dispersive propagation in the medium of interest.
- AC-003 (Exponential attenuation) is applicable only under: homogeneous lossy medium; single frequency.
- AC-001 (Acoustic impedance contrast (pulse-echo)) is applicable only under: normal incidence, plane wave, lossless media.
- Model input variables requiring sourced values before any numeric result: c (m/s), f (Hz), A_0 (-), alpha (1/m), d (m), Z_1 (Rayl), Z_2 (Rayl) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: lambda (m), A (-), R (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): operating frequency [f, Hz] value_status=UNKNOWN; element aperture [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; matching layer Z [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; excitation voltage and duty [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; detection threshold [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; coupling-path attenuation [alpha, 1/m] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'adverse event without identified device ', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'detection threshold missed at depth (attenuation)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'crosstalk between elements corrupting detection', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resonance drift with temperature', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'bond-wire fatigue under cyclic excitation', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coupling loss at interface deairing failure', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'pacemaker lead' must maintain function under the conditions of occurring injury events with re	SOURCE_FACT
DI-002	Mechanism driver (proposed)	A lifecycle-based risk management framework that integrates complication management across the entire continuum of pa	MODELLED
DI-003	Expected effect target	Early detection of lead complications before they result in adverse events with MODIFIED device problems	MODELLED
DI-004	Domain design input pattern: targeted applicability	NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: allowed applicability (KATE F. S. T. A. G. L. S. H. E. B. (c. 2011))	candidate pattern; confirm against ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: determined applicability	DEFINED via tokens ['detect', 'detection']	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards	ISO 60601-2-67 (in registry category UNKNOWN) (IEEE 730C PROPOSED) is never inferred	UNKNOWN
DI-008	Custodied observation: Long Term	https://andrepict.org/a/Release/1905594/defineAftera/342926426012021/PAGE4462afab24e8d54734d13	
DI-009	Custodied observation: Bradyarrhythmia	https://arxiv.org/abs/1804.09118 ; An Update R33e0c54b1020980ce5f63cc6dd1c64ed04028933c8f	
DI-010	Custodied observation: Preventing complications in pacemaker leads	https://arxiv.org/abs/1804.09118 ; An Update R33e0c54b1020980ce5f63cc6dd1c64ed04028933c8f	
DI-011	Custodied observation: Impact of patient factors on pacemaker lead performance	https://arxiv.org/abs/1804.09118 ; An Update R33e0c54b1020980ce5f63cc6dd1c64ed04028933c8f	
DI-012	Custodied observation: Simultaneous Transcatheter and Extracatheter Operative Episodes	https://arxiv.org/abs/1804.09118 ; An Update R33e0c54b1020980ce5f63cc6dd1c64ed04028933c8f	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	transducer element + matching layer realizing Implementation	CONCEPTUAL	transducer architecture defines acoustic sensing to detect lead dislodgement
DO-002	excitation / receive electronics realizing Implementation	CONCEPTUAL	system architecture defines acoustic sensing to detect lead dislodgement with
DO-003	signal chain (detection criterion) realizing Implementation	CONCEPTUAL	system architecture defines acoustic sensing to detect lead dislodgement with
DO-004	coupling path to target medium realizing Implementation	CONCEPTUAL	system architecture defines acoustic sensing to detect lead dislodgement with
DO-005	transducer element + matching layer interfaces with excitation / receive electronics	CONCEPTUAL	signal processing (all hands on deck) a comprehensive moni
DO-006	excitation / receive electronics interfaces with signal chain (detection criterion)	CONCEPTUAL	signal processing (all hands on deck) a comprehensive moni
DO-007	signal chain (detection criterion) interfaces with coupling path to target medium	CONCEPTUAL	signal processing (all hands on deck) a comprehensive moni
DO-008	Device boundaries: patient/fluid inlet and outlet termination	CONCEPTUAL	control/data points required by system seat definitions
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	half dimensions, tolerances, surface finish
DO-010	Operating range for operating frequency within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for operating frequency; verification of range
DO-011	Operating range for element aperture within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for element aperture; verification of range
DO-012	Operating range for matching layer Z within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for matching layer Z; verification of range
DO-013	Operating range for excitation voltage and duty within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for excitation voltage and duty; verification
DO-014	Operating range for detection threshold within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for detection threshold; verification of rang
DO-015	Operating range for coupling-path attenuation within which the mechanism holds its state	UNKNOWN	state behavior computed bounds for coupling-path attenuation; verification o
DO-016	No closed-loop control is proposed: the invention operates as an open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: impedance-contrast detection threshold	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation
DO-019	Fixture enabling: crosstalk and noise floor characterization	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation
DO-020	Fixture enabling: durability under cyclic excitation — rig concept named, build specifications	UNKNOWN	rig concept named, build specifications UNKNOWN; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: phantom detection-threshold mapping	UNKNOWN - rig concept framed, guidelines, instrumentation	UNKNOWN - CS selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
operating frequency	UNKNOWN (no sourced value)	Hz	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
element aperture	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
matching layer Z	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
excitation voltage and duty	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
detection threshold	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
coupling-path attenuation	UNKNOWN (no sourced value)	1/m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
adverse event without fail-safe device	Identified device event without identified device	Identified device	Identified device	Identified device
detection threshold miss	Failure depth (attenuation) appeared and its detection was not possible	Failure depth (attenuation) appeared and its detection was not possible	Failure depth (attenuation) appeared and its detection was not possible	Failure depth (attenuation) appeared and its detection was not possible
crosstalk between elements	Failure by crosstalk between elements	Failure by crosstalk between elements	Failure by crosstalk between elements	Failure by crosstalk between elements
resonance drift with temperature	Failure by resonance drift with temperature	Failure by resonance drift with temperature	Failure by resonance drift with temperature	Failure by resonance drift with temperature
bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation
coupling loss at interface	Failure by coupling loss at interface	Failure by coupling loss at interface	Failure by coupling loss at interface	Failure by coupling loss at interface

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
adverse event without identification	Failure by adverse event without identification	design controls (NOT ESTABLISHED)	design controls (NOT ESTABLISHED)
detection threshold missed	Failure by detection threshold missed at depth (at and frequency selection)	depth (at and frequency selection)	depth (at and frequency selection)
crosstalk between elements	Failure by crosstalk between elements	crosstalk (at and frequency selection)	crosstalk (at and frequency selection)
resonance drift with temperature	Failure by resonance drift with temperature	resonance drift with temperature	resonance drift with temperature
bond-wire fatigue under cyclic excitation	Failure by bond-wire fatigue under cyclic excitation	bond-wire fatigue under cyclic excitation	bond-wire fatigue under cyclic excitation
coupling loss at interface	Failure by coupling loss at interface	coupling loss at interface	coupling loss at interface

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Compare complication rates in patients	Compare complication rates in patients	Compare complication rates in patients	Compare complication rates in patients
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	impedance-contrast detection threshold	impedance-contrast detection threshold	impedance-contrast detection threshold	impedance-contrast detection threshold
VF-004	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping	phantom detection-threshold mapping
VF-005	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature	resonance tracking across the temperature
VF-006	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out	accelerated cyclic-excitation run-out
VF-007	transmitted-energy check with deliberate	transmitted-energy check with deliberate	transmitted-energy check with deliberate	transmitted-energy check with deliberate

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	PZT / PVDF active element	standard ultrasonic transducers	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
PZT / PVDF active element	domain candidate material	1	NOT ESTABLISHED	PZT / PVDF active element	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- diced-ceramic + backing assembly: domain registry manufacturing route - candidate only; no process qualification exists
- matching-layer lamination and bonding: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter operating frequency [f Hz]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter element aperture [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter matching layer Z [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter excitation voltage and duty [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter detection threshold [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter coupling-path attenuation [alpha 1/m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification frequency	impedance-contrast	NOT ESTABLISHED (to be quoted)	impedance-contrast	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED (to be quoted)	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Compare complication	NOT ESTABLISHED (to be quoted)	Compare complication	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED (to be quoted)	run killer experiment:	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	impedance-contrast	NOT ESTABLISHED (to be quoted)	impedance-contrast	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	phantom detection threshold	NOT ESTABLISHED (to be quoted)	phantom detection threshold	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	resonance tracking	NOT ESTABLISHED (to be quoted)	resonance tracking	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	accelerated cyclic-exposure	NOT ESTABLISHED (to be quoted)	accelerated cyclic-exposure	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	transmitted-energy	NOT ESTABLISHED (to be quoted)	transmitted-energy	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.